

Capture-The-Flag: Crypto
CS6903: Network Security
2025-26 - Semester II

Department of Computer Science and Engineering
Indian Institute of Technology Hyderabad

Start Time: 20th Feb 2026, 6:00pm

End Time: 22nd Feb 2026, 6:00pm [*No extension]

Writeup Due Date: 23rd Feb 2026, 9:00am [*No extension]

***Disclaimer:** This contest aims to familiarize you with some offensive techniques in computer security. You will be carrying out attacks with our permission in a controlled environment. Keep in mind that doing similar attacks on other machines without authorization is a legal offense. You may face disciplinary and legal action for unauthorized attacks.*

The take-home exam is in the format of a capture-the-flag security contest. You will be required to solve confidentiality-related problems, mostly by breaking crypto and extracting the plaintext from encrypted data. When you complete a challenge, you will get a flag, which is a passphrase that you have to submit to get points for that challenge.

Here are the rules for the contest:

- The contest has 12-15 challenges (or maybe more) in increasing order of difficulty, and more problems can be added during the CTF.
- **You must have a Linux machine to carry out the CTF tasks.**
- The main contest (CTF) server is available at <https://192.168.51.76>, where you have to submit your flags.
- The challenges are hosted on **192.168.51.198**. To solve these challenges, you have to SSH into this machine with the credentials mailed to you.
- Your task is to obtain the plaintext for a given problem, which will reveal the flag for that challenge. The source code for some problems is available for download from the CTF server, which was used to encrypt the plaintext.
- Teams are not allowed in this CTF. This is an individual event.
- All flags are of the format **cs6903{<passphrase>}**. For example, if you see **cs6903{supersecret}** on solving a challenge, submit **cs6903{supersecret}** on the contest page. It is important that you submit the flags as you capture them. Don't delay the submission.
- **You have a maximum of 5 attempts (submissions) per challenge. The attempt count will not increase if it is exhausted.**
- The contest will start at **6:00pm, Feb 20th, 2026**, and remain active till **6:00pm, Feb 22nd, 2026**, and will be accessible only from the **IITH Network or through IITH-VPN**.

- We will share the platform access credentials 5 minutes before the scheduled start time.
- Write-up submission is due by **9:00am, Feb 23rd, 2026**. No extension will be given.
- Teams/Collaboration is not allowed, as this is an individual event and not group-based. If any student is found to have shared their work with another student, both will receive 0 (zero) for this take-home, followed by a one-grade reduction for this course. For example, if the final grade for the course is B, it will be reduced to B-. Several monitoring mechanisms are in place to detect malpractice.
- If you feel lost or need clarification, feel free to ask the TAs and the Course Instructor.
- Any attempts to DoS the contest infrastructure will attract penalties.
- If you find any weakness in any of the challenges, let us know without trying to misuse it. You may be awarded bonus points for responsible disclosure.

Deliverables: You should submit a zip file containing subdirectories with the name of the problem. Each subdirectory should have a file `answer.txt` describing, in not more than 5 lines, how you solved or attempted to solve the challenge. If you wrote any small code/script for that challenge, include it in the subdirectory. Include a file `'it.txt'` in the root directory mentioning the maximum penalty under the Indian IT Act, 2000 for "Hacking with computer system" and "Securing access or attempting to secure access to a protected system."

Submission Format:

- Submit a **tar-ball/zip** file of your CTF assignment named as your **roll number**. For example, if your roll number is CSXXMTECH12345, then the tar/zip file name will be CSXXMTECH12345.tar.gz or CSXXMTECH12345.zip.
- The zip file should contain the data as described in the deliverables.
- Upload your submission to Moodle. Submissions via email or any other medium are strictly prohibited.

All the best. Looking forward to the submissions !!.....PS: Start Early.